AI-based Solution for Learning Disabilities

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Student Declaration

We hereby declare that the work presented in the report entitled **"AI-based Solution for Learning Disabilities"** submitted by us for the partial fulfilment of the requirements for the degree of B.Tech. at Indraprastha Institute of Information Technology, Delhi, is an authentic record of our work carried out under the guidance of **Prof. Md. Shad Akhtar** , **Co-advisor: Mr. Aditya Bhardwaj**. Due acknowledgements have been given in the report for all material used. This work has not been submitted elsewhere for the reward of any other degree.

Aarushi Verma & Aayush Gupta **Place & Date:** IIIT Delhi, November 28, 2025

Certificate

This is to certify that the above statement made by the candidates is correct to the best of our knowledge.

Prof. Md. Shad Akhtar
Co-advisor: Mr. Aditya Bhardwaj

Place & Date: IIIT Delhi, November 28, 2025

Abstract

Dyslexia affects a large number of students in India, yet most learning-support technologies are built using Western datasets and reading behaviors. The absence of native multimodal datasets limits the development of tools that can understand the linguistic, cognitive, and cultural patterns of Indian dyslexic learners.

This project proposes a web-based, multimodal data collection platform that gathers reading, writing, speech, mathematical comprehension, and typing patterns from school-age children in India. The portal supports role-specific workflows for schools, parents, students, and administrators to ensure controlled, verified, and ethical participation.

The platform includes gamification (badges, scores, statistics), accessibility support (dyslexic-friendly UI), and consent-based participation. The dataset collected through our system will serve as a foundation for training future models in automated dyslexia screening, error analysis, assistive reading platforms, and personalized learning systems, tailored specifically for Indian students. The current phase focuses on data acquisition, ethical compliance, and platform deployment for school-level trials.

Keywords: Dyslexia, Educational Technology, Data Collection, Multimodal Dataset, Indian Languages, Accessibility, Gamification.

Acknowledgment

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Work Distribution

Student	Responsibilities
Aarushi Verma	**Parent Platform, Student Platform, Accessibility Suite** • **Parent Platform**: Engineered the secure Parent Dashboard to serve as the primary consent gatekeeper, implementing workflows for independent registration, child account creation, and longitudinal progress visualization. • **Student Platform**: Developed the interactive Student Dashboard and task interfaces, integrating frontend logic for gamification (badges, scoring), task navigation, and real-time state management. • **Accessibility Suite**: Architected the custom DOM manipulation engine for the accessibility toolbar, creating features like the interactive reading ruler, OpenDyslexic font toggles, and high-contrast monochrome modes.
Aayush Gupta	**Admin Platform, School Platform, API Architecture** • **Admin Platform**: Built the centralized Admin Panel to monitor global system health, aggregating data to visualize cumulative statistics and participation trends across all partner schools. • **School Platform**: Developed the institutional School Dashboard, enabling granular class-wise management, bulk student/parent onboarding, and classroom-level performance tracking. • **API Architecture**: Architected the robust backend infrastructure using **Flask and Python**, designing custom API endpoints to handle complex role-based access control and high-concurrency data traffic.
Joint Work	**System Design, Database & Deployment** • **Database Schema**: Collaboratively structured the relational database to support the interconnected hierarchical relationships (School → Class → Student → Parent) while ensuring data anonymity. • **Conceptualization**: Collaboratively formulated the "Give and Take" platform strategy to incentivize data collection. • **Task Design**: Jointly designed the cognitive battery (Reading, Writing, Comprehension) based on literature gaps. • **UI/UX Design**: Lead the design of the responsive interface, ensuring an intuitive user journey for non-technical parents and students. • **Cloud & Deployment**: Worked together on configuring the University Server deployment and setting up cloud storage buckets for audio/image assets. • **Testing**: Conducted end-to-end testing and gathered initial feedback from partner schools to iterate on the workflow.

Contents

Chapter 1

Introduction

1.1 Background

Dyslexia is a neurodevelopmental learning disorder affecting reading, writing, and comprehension capabilities. Despite large literacy challenges in India, most datasets used to support dyslexia research and model development originate from Western languages, predominantly English. These datasets do not fully represent phonetic structures, multilingual environments, or classroom patterns in Indian schools.

1.1.1 Motivation

The lack of Indian-context data restricts the development of effective learning tools, while schools and parents currently lack accessible assessment platforms. Recognizing that voluntary participation is unsustainable without incentives, we pivoted from a simple data submission portal to a comprehensive educational platform. By adopting a reciprocal "Give and Take" model offering users tangible feedback through scores, stats, and progress tracking - we incentivize participation. This approach ensures ethical, validated data collection while simultaneously providing essential assessment infrastructure to the Indian educational ecosystem.

1.1.2 Problem Statement

There is no existing multimodal dyslexia dataset representing Indian students, collected through ethically-verified, school-supported, task-based interactions, that can enable model training for screening and personalized learning.

1.1.3 Objective

- To build a secure, validated, multimodal dyslexia dataset collection platform.
- To collect reading, writing, speech, typing, arithmetic comprehension, and aptitude responses.

- To integrate accessibility and gamification for comfort and motivation.
- To enable future AI models tailored to Indian learners.

Chapter 2

Literature Review

Research studies indicate that multimodal behavioral cues such as speech errors, orthographic patterns, reading pace, comprehension ability, eye-tracking, and keystroke behavior - help improve the accuracy of dyslexia screening and learning support systems. Existing works highlight that analyzing audio pronunciation, handwriting patterns, and typing hesitation can provide meaningful indicators of reading disability. These findings suggest that multimodal datasets offer richer features than text-only screening assessments traditionally used in schools.

Recent advancements in Artificial Intelligence have further validated this multimodal approach. A 2025 scoping review by Yap et al. highlights that AI integration in dyslexia research is shifting towards personalized learning interventions and early detection using diverse data points, yet notes a significant gap in non-English tools [4]. Similarly, Zhao et al. demonstrated that AI-driven reading assistants can significantly enhance reading abilities, but their success relies heavily on the availability of high-quality, diverse training data [5].

However, current dyslexia datasets and screening tools face several limitations:

- **Lack of Indian languages & reading curricula:** Majority of datasets are based on English, whereas Indian learners navigate multilingual contexts and phonetic systems.
- **Absence of parent-assisted consent workflows:** Ethical participation, especially with minors, requires structured parental or guardian involvement.
- **Limited multimodal data collection at scale:** Existing resources are either clinically restricted, small-scale, or lack varied modalities (speech, handwriting, typing). A review of mobile applications by Molina Villamarín et al. emphasizes that while mobile strategies improve literacy, they often lack integration with formal educational structures [1].
- **No ethics-first web platforms for schools:** Few systems integrate secure, verified student participation with cultural and institutional requirements specific to Indian education.

Inspired by ethical and workflow principles and recent frameworks for integrating Large Language Models (LLMs) into therapeutic education [2], our platform integrates:

- **Secure Login & Identity Validation** to ensure controlled school-driven participation.
- **Consent Tracking** with parent and school authorization before participation.
- **Gamified Task Interfaces** designed to reduce reading anxiety and improve student engagement.
- **Post-Collection Annotation Modules** enabling experts to manually refine multimodal labels for research-grade datasets.

Thus, while prior research establishes the value of multimodal dyslexia cues, our work uniquely addresses the Indian educational context through *scalable, ethically verified, web-based, multimodal data collection tailored for children.*

Chapter 3

Methodology

This project follows a systems-engineering approach to design, implement, and validate a **standardized, multimodal data-collection platform** for Indian students with dyslexia. The emphasis of this semester was on systems engineering: workflow modeling, backend architecture, database design, frontend development, task engine construction, accessibility tooling, and deployment preparation. The machine learning and analysis pipelines will be undertaken in next semester.

3.1 System Architecture Overview

The platform is implemented as a modular, role-driven web system built on a Flask backend, MySQL storage layer, and server-rendered interfaces augmented with client-side JavaScript. The overall system follows a four-layer architecture:

- **Backend Layer (Flask and Python)** : Handles all routing, REST-like API endpoints, task submission, RBAC enforcement, and upload handling. Templates are rendered using Jinja2, and each user role is mapped to dedicated routes.
- **Frontend Layer (HTML/CSS/JS)** : Implements dynamic dashboards for School, Parent, Student, and Admin users using HTML templates with JavaScript for state management, timers, audio recording, validation, and save/submit flows.
- **Storage Layer (MySQL and File Storage)** : Stores user profiles, school/class mappings, consent, demographics, task definitions, attempts, and progress. Media files (audio and uploads) are stored in a protected `uploads/` directly.
- **Authorization Layer** : A custom role-based access control (RBAC) system ensures that each request is authenticated, authorized, and context-validated before any database access or task visibility.

This architecture enables controlled, secure, and research-grade multimodal data acquisition at scale.

3.2 System Roles

The platform supports role-based workflows to ensure ethical, secure, and validated multimodal data collection. Table 3.1 summarizes the functionalities associated with each role.

Role	Key Features
School	Class creation, bulk student listing (CSV or manual), linking students to parents, enabling participation, viewing school-wide statistics and task completion heatmaps
Parent	Independent signup, child registration, demographic submission, consent authorization, viewing detailed progress and task scores, launching student portal
Student	Accesses student portal via parent, performs multimodal tasks, saves progress, submits attempts, retakes tasks, views badges, scores, and analytics
Admin	Monitors platform-wide statistics, manages schools, views global participation analytics, audits data quality, and oversees system health

Table 3.1: System roles and associated privileges.

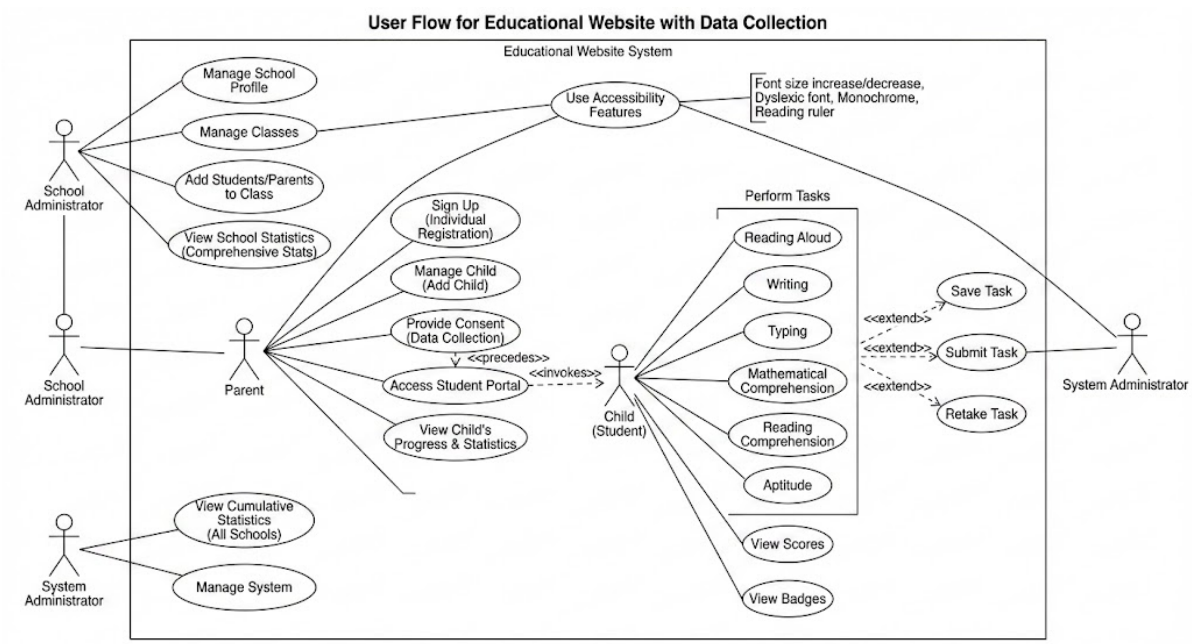

Figure 3.1: High-level system architecture illustrating the interactions between School, Parent, Student, and Admin workflows along with backend and database components.

3.3 Backend Engineering & API Architecture

A major engineering component of this semester was building a robust backend to support high-concurrency usage by schools. Key components include:

- **RBAC Middleware:** Every route validates the user's role (School, Parent, Student, Admin) and checks that the user has ownership of school/class/child records.
- **Task Routing Engine:** Maps each student to a canonical set of tasks, manages unique attempt creation, enforces save/submit/retake logic, and timestamps all attempts.
- **File Upload Management:** Audio and file submissions are validated for type/size, stored under randomized filenames, and referenced in the database without exposing paths.
- **Consent & Demographic Pipeline:** Tasks remain locked until parent consent and demographics are submitted. Demographic triggers automatically compute age from date of birth.
- **Scoring Engine:** Each progress table stores `score` and `max_score`, enabling normalized scoring for future ML pipelines.

3.4 Database Design

The MySQL schema is normalized, hierarchical, and engineered to ensure data integrity, provenance, and scalability during large-scale multimodal data collection. The database is organized into entity tables, task-definition tables, attempt provenance tables, and task-specific progress tables.

3.4.1 Core Schema Components

- **Entity Tables:** `users`, `schools`, `classes`, `consent_data`, and `demographics` store the core institutional hierarchy, participant information, consent status, and demographic metadata.
- **Task Definitions:** `tasks` and `reading_tasks` define the canonical structure of each task. Additional SQL migrations (e.g., aptitude and comprehension task expansions) allow the schema to evolve without disrupting existing data.
- **Attempt Provenance:** `user_tasks` maps each user to their assigned tasks, while `user_task_attempts` stores every saved or submitted attempt along with timestamps, attempt mode (save/submit), and status.
- **Progress Tables:** `typing_progress`, `comprehension_progress`, `mathematical_comprehension_progress`, `aptitude_progress`, and `writing_samples` capture task-specific scoring, response structures, and derived metrics.

3.4.2 Data Integrity and Relational Logic

The schema incorporates several mechanisms to ensure correctness and consistency:

- **Foreign Key Constraints:** All major tables enforce referential integrity. Deleting a school or class cascades to dependent entries, preventing orphaned rows.

- **Unique Constraints:** The platform enforces a `UNIQUE(user_id, task_id)` constraint in `user_tasks`, ensuring that each student receives exactly one canonical mapping to each task. All retakes are stored under the same mapping via multiple attempts.
- **Attempt Logic:** Each attempt is associated with a single `attempt_id`. Save events create or update a partial attempt state; Submit events finalize the attempt and write to the corresponding progress table.
- **Separation of PII and Task Data:** Personally identifiable information is stored only in `users` and `demographics`. Task responses and uploads reference users only via foreign keys, enabling pseudonymized exports for later works.
- **File Integrity:** Uploaded audio/text files are stored outside the webroot with randomized filenames. The database stores only the reference path, protecting file integrity and preventing direct public access.

This design ensures that every data point is traceable, validated, ethically compliant, and compatible with downstream machine learning workflows.

3.5 Task Design

Multimodal tasks are designed based on **Bloom Taxonomy** to capture linguistic, cognitive, and behavioral indicators relevant to dyslexia screening. Table 3.2 lists tasks and corresponding collected data modalities.

Task	**Data Collected**
Reading Aloud	Audio via Web Audio API, pronunciation cues, reading pace, pause durations
Writing	Typed text, keystroke logs, structural errors, insertion/deletion events
Typing	Millisecond-level keystroke timing, speed, reversal patterns, latency bursts
Reading Comprehension	MCQ responses, time per question, behavioral pause detection
Mathematical Comprehension	Error propagation patterns, numeric mistakes, reasoning flow
Aptitude	Cognitive strategies, logical reasoning accuracy, question-level timing

Table 3.2: Tasks and multimodal data collected.

Each task supports saving partial progress, final submission, and full retakes to improve data richness.

3.6 Frontend Engineering & User Experience

The user interface combines Jinja2-rendered templates with JavaScript-driven interactivity. Engineering contributions include:

- **Dynamic Dashboards:** School, Parent, Student, and Admin dashboards implemented with responsive design and conditional rendering based on user role.
- **Task Interfaces:** Timers, audio recorders, validation prompts, reading passages, MCQs, and interactive writing editors.
- **State Management:** Save-and-resume logic implemented via AJAX calls, enabling students to complete tasks across sessions.
- **Gamification Components:** Progress indicators, milestone badges, and task completeness summaries.

This ensures a highly engaging and stable user experience across laptops, desktops, and tablets.

3.7 Accessibility Integrations

To ensure comfort for students with learning difficulties, the platform integrates accessibility features including:

- Dyslexic font toggle
- Reading ruler overlay
- Monochrome mode to reduce visual strain
- Adjustable font sizes for readability

All accessibility tools are implemented client-side using DOM manipulation scripts to ensure performance and zero backend overhead.

3.8 Gamification

Gamification is incorporated to reduce reading anxiety and increase task engagement through:

- Progress bars to indicate task completion
- Reward badges for participation and improvement
- Score tracking with student dashboard statistics

- Reattempt options to encourage low-pressure learning

The scoring system uses normalized `score/max_score` fields, enabling cross-task comparison during analysis.

3.9 Ethical Compliance

The platform prioritizes child safety, data protection, and transparent consent via:

- IRB-aligned digital consent mechanisms
- Demographic information stored in anonymized form
- Parent-verified participation before data collection item Pseudonymization-ready schema for downstream dataset release

All data flows are designed to support later pseudonymization and controlled dataset export for research use later.

3.10 Testing and Deployment Preparation

To validate the functionality and usability of the platform, we conducted a small pilot study amongst our peers. This study enabled us to test the complete end-to-end workflow, including role-based access, task flows, save/submit/retake logic, file uploads, and the accessibility suite. Feedback from this pilot informed interface refinements, error-handling improvements, and adjustments to scoring and task presentation.

The platform is now stable for larger-scale deployment soon with actual schools.

Chapter 4

Results

This chapter summarizes the technical outcomes of this semester, during which the primary objective was to engineer a fully functional, role-based multimodal data-collection platform. The results reflect the completion of core backend systems, database infrastructure, task pipelines, user workflows, and initial stability validation through a controlled pilot study.

4.1 Platform Functionality Achieved

By the end of this semester, the critical platform components required for standardized multimodal data collection were fully implemented:

- **Backend Completion:** All Flask routes for authentication, role-based access control, consent handling, demographic capture, task assignment, scoring, and file uploads were implemented and verified. The backend enforces strict contextual checks (school → class → parent → student) to maintain data integrity across workflows.
- **Database Stability:** The MySQL schema, comprising entity tables, task definitions, attempt provenance, and progress tables, proved stable under repeated insertion, update, and retake operations. Constraints, foreign keys, and triggers behaved as intended, with no integrity violations observed during testing.
- **Task Engine Implementation:** All six multimodal tasks were deployed with complete interaction logic, including timers, audio recording, keystroke capture, MCQ rendering, scoring, and structured metadata logging. The save, submit, retake pipeline functioned consistently across task types.
- **User Dashboards:** Dedicated dashboards for School, Parent, Student, and Admin roles were fully integrated with live backend data. Real-time task status, completion states, and scoring feedback rendered correctly for all participants.
- **Accessibility Suite:** Core accessibility features (dyslexic font, monochrome mode, reading ruler, font resizing) performed reliably across all pages without breaking layout or task flows.

4.2 Data Pipeline Outcomes

The complete data pipeline, from participant creation to multimodal submission, was validated, confirming that:

- Each new task attempt generates a unique attempt ID in `user_task_attempts`.
- File uploads (audio recordings, typed samples) are stored under randomized filenames and correctly referenced in the database.
- Progress tables (`typing_progress`, `comprehension_progress`, etc.) receive correctly normalized `score` and `max_score` entries for each submitted attempt.
- Retakes do not overwrite earlier attempts; all attempt histories remain traceable for future model training.
- Metadata such as timestamps, attempt type, and recorded duration are consistently logged for downstream analysis.

These results indicate that the engineered schema and backend logic are ready for high-volume data collection in the next semester.

4.3 Pilot Study and Functional Validation

A small pilot study was conducted among peers to test the end-to-end workflow under realistic usage patterns. Key findings include:

- **Workflow Integrity:** School → Parent → Student transitions performed without broken links or permission failures. Consent gating effectively blocked task access until authorization was provided.
- **Task Execution:** Participants were able to complete all six tasks without encountering submission errors. Audio recordings saved consistently across different device microphones, and keystroke logs for typing and writing tasks captured accurate timing information.
- **Backend Robustness:** Multiple concurrent attempts did not generate duplicate user–task mappings or inconsistent scoring entries. The RBAC layer correctly prevented unauthorized access to restricted views.
- **User Experience Feedback:** The accessibility toolbar improved readability for several testers. Save-and-resume functionality behaved as expected, making longer tasks more user-friendly.

The pilot confirmed that the platform is reliable enough for real school deployments and requires no structural redesign.

4.4 Summary

This semester we successfully delivered a fully operational, technically stable platform capable of collecting multimodal data in a controlled, validated, and scalable manner. All core engineering systems, backend, database, dashboards, task engine, scoring, accessibility, and integrity pipelines, were completed and verified through practical testing. The system is now prepared for large-scale participant onboarding and dataset generation in the next semester.

Chapter 5

Future Work

Future enhancements are planned to strengthen scalability, research usability, and linguistic diversity of the platform. The proposed developments include:

- **Deployment at partner schools** to enable large-scale, real-world multimodal data acquisition.
- **Annotator dashboard for linguists and clinicians** to support expert-driven labeling and data correction.
- **Speech model integration for immediate feedback** enabling automated pronunciation assessment and screening support.
- **Multilingual expansion (Hindi and regional languages)** to reflect India's linguistic diversity in dyslexia learning contexts.
- **Dataset release for the research community** under ethical and consent-based licenses to encourage development of assistive learning models.

Chapter 6

Conclusion

This semester established the complete technical foundation for a standardized, ethical, and scalable multimodal data-collection platform for Indian dyslexic learners. We engineered a fully functional system with role-based workflows, a secure Flask backend, a normalized MySQL schema, stable attempt-tracking logic, and comprehensive support for all six multimodal tasks. The platform integrates accessibility features, consent-driven participation, and gamified feedback, ensuring usability for students, parents, and schools.

A pilot study among peers validated the correctness of the workflows, robustness of the backend, reliability of file handling, and stability of the task engine. These results confirm that the platform is ready for real-world deployment in schools.

Overall, Semester 1 successfully delivered a reliable, secure, and extensible system that meets the engineering requirements for large-scale data acquisition. Semester 2 will build on this foundation through actual data collection, dataset preparation, and the development of machine-learning models tailored to Indian dyslexic learners.

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